

TPMS Software Library Release Notes

MLX91804

Abstract

This document describes the contents of the TPMS Software Library release.

History records

Date	Version	Comments
Jan 10, 2018	1.7.1	Obsolete version. Not recommended for the new designs.
Jul 10, 2018	1.8.1	Obsolete version. Not recommended for the new designs. Added support for new 91804CA devices; updated LF bootloader
Jul 24, 2018	1.8.2	Obsolete version. Not recommended for the new designs. Fixed a bug in the LF bootloader with a wrong stack allocation
Jun 6, 2019	1.9.0	Obsolete version. Not recommended for the new designs. Improves library safety, increases its performance and simplifies user experience
Jan 10, 2020	1.10.0	Supports prebuilt profile modules to optimize the library integrity and compilation time; improved RFTx and LFRx modules; added I2C module as well as the pressure, acceleration and memory diagnosis.
Oct 16, 2020	1.10.3	Added new features: memory data validity check, oscillators' calibration with the external reference frequency, I2C stretching and the conversions based on the acceleration sensing compensation; updated LFRx configuration; improved the functions RfTx_Run and RfTx_RunFromBuf; corrected the I2C module and the function Check_IsBadPressure.
Aug 20, 2021	1.10.5	LFRX gain setting was upgraded, added new function returning an extended chip ID, examples were updated.
May 5, 2022	1.10.6	Premain configuration at cold bootstrap was fixed to ensure stable wake-ups at the low-temperature range. LFRx factory trimming was extended. The function Check_IsBadPressure was improved.
Feb 29, 2024	1.10.8	Phys_AccelerationX/Z functions were optimized. Added a new function recognizing the input pressure range.
May 24, 2024	1.10.9	The calibration algorithm in the function RfTx_Calibrate was improved.
Nov 12, 2024	1.10.11	ISR_ADC_XH018 in acceleration measurements with hardware filtering was updated. The function NvRam_Recall was improved.

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1. Release contents

The TPMS library package provides convenient interface to the following features of the TPMS chip (<name> below means a right side of the different functions names):

- **AbsWatchdog_<name>**: Absolute Watchdog timer (AWD) Interface, <sensorlib.h>
- **Accel_<name>**: Raw Acceleration measurement conversions, <sensorlib.h>
- **Accel_<name>**: Raw Acceleration measurements with Low Pass HW Filter, <sensorlib.h>
- **Adc_<name>**: Analog-to-digital conversions to get raw Pressure, raw Temperature, etc., <sensorlib.h>
- **Atomic sections support**, <atomic.h>
- **builtin_<name>**: Builtin MLX16 CPU functions, <builtin_mlx16.h>
- **Calc_<name>**: PVT sensors' compensation and memory signature computation, <sensorlib.h>
- **Check_<name>**: Sensors' operability and nvRAM factory data validation, <sensorlib.h>
- **Diag_<name>**: Diagnostic Interface support, <diag.h>
- **DEBUG_<name>**: Debug functions, <debug.h>
- **Flash_<name>**: Flash memory Interface, <sensorlib.h>
- **Gpio_<name>**: General-purpose input/output pins, <sensorlib.h>
- **I2C_<name>**: I2C master functions, <i2c_master.h>, <i2c_master_trx.h> (Obsolete module)
- **Itc_<name>**: Interrupt Controller (ITC) Interface, <sensorlib.h>
- **Lf_<name>**: Low Frequency (125 kHz) Transceiver common functions, <sensorlib.h>
- **LfRx_<name>**: Low Frequency (125 kHz) Receiver Interface, <sensorlib.h>
- **LfTx_<name>**: Low Frequency (125 kHz) Transmitter Interface, <lftx.h>
- **NvRam_<name>**: Non-volatile random-access memory (nvRAM) Interface, <sensorlib.h>
- **Phys_<name>**: Acceleration measurement in physical units, <sensorlib.h>
- **RfTx_<name>**, **Radio_<name>**: Radio frequency (300-450 MHz) Transmitter Interface, <sensorlib.h>
- **SimpleTimer1_<name>**, **SimpleTimer2_<name>**: Simple Timers' Interface, <sensorlib.h>
- **Sys_<name>**: System Library Functions, <sensorlib.h>
- **TempSensor_<name>**: Temperature Sensor Interface, <sensorlib.h>
- **Wakeup_<name>**: Wake-up Controller Interface, <sensorlib.h>
- **WakeupTimer_<name>**: Wake-up Timer Interface, <sensorlib.h>

2. Release History

2.1 Version 1.10.11

The library supports the same set of chip variants as the previous library versions 1.10.9, 1.10.8, 1.10.6, 1.10.5 and 1.10.3 (see the next chapters for details).

2.1.1 Fixed issues

- The **ISR_ADC_XH018** in acceleration measurements with hardware filtering block has been updated to prevent corruption of the **Phys_AccelerationX** and **Phys_AccelerationZ** readout data.
- The function **NvRam_Recall** was upgraded to work stably at a low VDD supply range.

2.2 Version 1.10.9

The library supports the same set of chip variants as the previous library versions 1.10.8, 1.10.6, 1.10.5 and 1.10.3 (see the next chapters for details).

2.2.1 Fixed issues

- The calibration algorithm of the functions **RfTx_Calibrate** was updated to ensure the better accuracy.

2.3 Version 1.10.8

The library supports the same set of chip variants as the previous library versions 1.10.6, 1.10.5 and 1.10.3 (see the next chapters for details).

2.3.1 Fixed issues

- The functions **Phys_AccelerationX** and **Phys_AccelerationZ** were updated to optimise the output noise and power consumption.
- The function **Check_IsBadPressure** was updated to support all produced devices over the operating conditions.

2.3.2 New features

- The new function **Sys_GetPressureRange** is added to define the input pressure range of the device. The new software example `pressure_sensor_range` is added to show how to use the new function.

2.4 Version 1.10.6

The library supports the same set of chip variants as the previous library versions 1.10.3 and 1.10.5 (see the next chapters for details).

2.4.1 Fixed issues

- The function **Check_IsBadPressure** was updated to optimise its power consumption.

- Added a new configuration of the sleep modes. The configuration is executed at the cold bootstrap procedure to enable the register `VDIG_SHORT = 1`. It keeps the internal `VDIG` voltage pulled down to ensure stable wake-ups at low temperatures.

2.4.2 Changes

- The register `LF_AGC_TRIM` is now loaded at the cold bootstrap from the `nvRAM` data. It will allow a chip dependent factory adjustment of the `LFRx` performance.

2.5 Version 1.10.5

The library supports the same set of chip variants as the previous library version 1.10.3 (see the next chapter for details).

2.5.1 New features

- The new function **`Sys_GetChipIdHi`** can be used to return high 16-bits of the full 48-bits chip ID stored in `nvRAM`.
- The example `i2c_master_fram` is added to be used as an alternative for the `I2C` software module that becomes deprecated starting from this library revision.

2.5.2 Changes

- The number of available values for the configuration parameter `LF_GAIN_SEL` are extended to three: `LF_GAIN_MIN`, `LF_GAIN_MID` and `LF_GAIN_MAX`. The details are available in the description of the function **`Lf_Init`**.
- To prevent confusion 4 examples are renamed: `diag_accel` to `accel_sensor_diagnostics`, `diag_pressure` to `pressure_sensor_diagnostics`, `diag_vdd` to `diag_interface` and `diag_vdd_in_deep_sleep` to `diag_interface_with_deep_sleep`.
- The functions supporting software `I2C` master are reconfigured to work with the internal weak pull-up resistors on both `SDA` and `SCL` lines, although it doesn't make the external pull-up resistors unnecessary. Note that the `I2C` software module becomes deprecated starting from this library revision. It is recommended to replace it with the source code accepted from the example `i2c_master_fram`.

2.5.3 Removed features

- The functions deprecated in the previous library versions are removed. See table below for details:

Removed deprecated function	Suggested replacement
<code>Accel_ToAduX</code>	<code>Accel_GToRawX</code>
<code>Accel_ToAduZ</code>	<code>Accel_GToRawZ</code>
<code>Adc_ClearErrors</code>	<code>Adc_ClearAllErrors</code>
<code>Adc_GetErrors</code>	<code>Adc_IsError</code>
<code>Gpio_WriteOpenDrainPin</code>	n/a
<code>Gpio_ToggleOpenDrainPin</code>	n/a
<code>I2c_SwIntRequest</code>	n/a
<code>NvRam_IsBusy</code>	n/a
<code>NvRam_StoreNoWait</code>	<code>NvRam_Store</code>
<code>NvRam_RecallNoWait</code>	<code>NvRam_Recall</code>
<code>Sys_IsWarmBoot</code>	<code>Sys_IsColdBoot</code> with inverted output logic
<code>TempSensor_Init</code>	<code>TempSensor_SetMode</code>
<code>Wakeup_Init</code>	<code>Wakeup_Enable</code> and <code>Wakeup_Disable</code>

2.6 Version 1.10.3

The library supports chip variants with different **pressure ranges** (100-500 kPa, 100-900 kPa and 100-1400 kPa) and different **acceleration sensing** (X, Z, XZ-axes or without acceleration sensor). Three variants of input ranges are available for the options with a single sensitive acceleration axis. The required chip variant has to be selected in the application's Makefile by assigning the **PROFILE** variable by one of the profiles names listed below:

PROFILE	Description
DEFAULT	Alias for 91804CA_9BAR_XZ; should not be used for application
91803BC_5BAR	BC: 100-500 kPa without acceleration sensor
91803BC_9BAR	BC: 100-900 kPa without acceleration sensor
91803BC_14BAR	BC: 100-1400 kPa without acceleration sensor
91803CA_5BAR	CA: 100-500 kPa without acceleration sensor
91803CA_9BAR	CA: 100-900 kPa without acceleration sensor
91803CA_14BAR	CA: 100-1400 kPa without acceleration sensor
91804BC_5BAR_X	BC: 100-500 kPa, X-acceleration sensor with symmetrical input range
91804BC_5BAR_X_NEG	BC: 100-500 kPa, X-acceleration sensor with asymm negative input range
91804BC_5BAR_X_POS	BC: 100-500 kPa, X-acceleration sensor with asymm positive input range
91804BC_5BAR_Z	BC: 100-500 kPa, Z-acceleration sensor with symmetrical input range
91804BC_5BAR_Z_NEG	BC: 100-500 kPa, Z-acceleration sensor with asymm negative input range
91804BC_5BAR_Z_POS	BC: 100-500 kPa, Z-acceleration sensor with asymm positive input range
91804BC_9BAR_X	BC: 100-900 kPa, X-acceleration sensor with symmetrical input range
91804BC_9BAR_X_NEG	BC: 100-900 kPa, X-acceleration sensor with asymm negative input range
91804BC_9BAR_X_POS	BC: 100-900 kPa, X-acceleration sensor with asymm positive input range
91804BC_9BAR_Z	BC: 100-900 kPa, Z-acceleration sensor with symmetrical input range
91804BC_9BAR_Z_NEG	BC: 100-900 kPa, Z-acceleration sensor with asymm negative input range
91804BC_9BAR_Z_POS	BC: 100-900 kPa, Z-acceleration sensor with asymm positive input range
91804BC_14BAR_Z	BC: 100-1400 kPa, Z-acceleration sensor with symmetrical input range
91804BC_14BAR_Z_NEG	BC: 100-1400 kPa, Z-acceleration sensor with asymm negative input range
91804BC_14BAR_Z_POS	BC: 100-1400 kPa, Z-acceleration sensor with asymm positive input range
91804CA_5BAR_X	CA: 100-500 kPa, X-acceleration sensor with symmetrical input range
91804CA_5BAR_X_NEG	CA: 100-500 kPa, X-acceleration sensor with asymm negative input range
91804CA_5BAR_X_POS	CA: 100-500 kPa, X-acceleration sensor with asymm positive input range
91804CA_5BAR_Z	CA: 100-500 kPa, Z-acceleration sensor with symmetrical input range
91804CA_5BAR_Z_NEG	CA: 100-500 kPa, Z-acceleration sensor with asymm negative input range
91804CA_5BAR_Z_POS	CA: 100-500 kPa, Z-acceleration sensor with asymm positive input range
91804CA_5BAR_XZ	CA: 100-500 kPa, XZ-acceleration sensor with symmetrical input range
91804CA_9BAR_X	CA: 100-900 kPa, X-acceleration sensor with symmetrical input range
91804CA_9BAR_X_NEG	CA: 100-900 kPa, X-acceleration sensor with asymm negative input range
91804CA_9BAR_X_POS	CA: 100-900 kPa, X-acceleration sensor with asymm positive input range
91804CA_9BAR_Z	CA: 100-900 kPa, Z-acceleration sensor with symmetrical input range
91804CA_9BAR_Z_NEG	CA: 100-900 kPa, Z-acceleration sensor with asymm negative input range
91804CA_9BAR_Z_POS	CA: 100-900 kPa, Z-acceleration sensor with asymm positive input range
91804CA_9BAR_XZ	CA: 100-900 kPa, XZ-acceleration sensor with symmetrical input range
91804CA_14BAR_X	CA: 100-1400 kPa, X-acceleration sensor with symmetrical input range
91804CA_14BAR_X_NEG	CA: 100-1400 kPa, X-acceleration sensor with asymm negative input range
91804CA_14BAR_X_POS	CA: 100-1400 kPa, X-acceleration sensor with asymm positive input range
91804CA_14BAR_Z	CA: 100-1400 kPa, Z-acceleration sensor with symmetrical input range
91804CA_14BAR_Z_NEG	CA: 100-1400 kPa, Z-acceleration sensor with asymm negative input range
91804CA_14BAR_Z_POS	CA: 100-1400 kPa, Z-acceleration sensor with asymm positive input range
91804CA_14BAR_XZ	CA: 100-1400 kPa, XZ-acceleration sensor with symmetrical input range

For example, to build an application with the 91804CA_9BAR_X profile, the Makefile should include the line:

```
PROFILE = 91804CA_9BAR_X
```

The DEFAULT profile is to simplify switching all library examples to any required profile, because all examples' makefiles include the line `PROFILE = DEFAULT`. By default the DEFAULT profile is an alias for the 91804CA_9BAR_XZ profile. It is not recommended to build an application with the DEFAULT profile.

The RFTx output power is set to 8 dBm for all profiles supporting 100-1400 kPa pressure range and 5 dBm for all other profiles. Application can reconfigure the default RFTx output power by **RfTx_Init** function taking **RfTxConfig_Type** configuration data with the **.extra** section (please see HTML library manual and **app_biphase_encoding** example for reference).

2.6.1 Fixed issues

- The function **Check_IsBadPressure** is corrected.
- I2C headers were added to the library package.
- At the initialization stage of the functions **RfTx_Run** and **RfTx_RunFromBuf** the crystal oscillator is not stopped being enabled before by the function **RfTx_Calibrate**. It optimized power consumption.

2.6.2 New features

- The function **RfTx_Run** is upgraded to return an error in case of an abnormal RFTx stop.
- The new function **Calc_RomBistFor** computes a 24-bit signature for the configurable memory range. The signature is based on the polynomial: $x^{24} + x^{23} + x^{22} + x^{17}$. The calculation is implemented by the ROM BIST hardware that reads the assigned memory data by means of the DMA.
- The new function **Check_IsBadMlxData** verifies integrity of the Melexis calibration and tracking data stored in the highest half of the nvRAM. The verification is based on the ROM BIST 24-bit signature calculation according to the polynomial: $x^{24} + x^{23} + x^{22} + x^{17}$.
- The I2C master functions are extended to support a clock synchronization (stretching) mechanism to slow down the clock speed on request from the I2C slave device.
- The new function **Sys_CalibrateHfoByExtClk** allows calibrating 8 MHz HFO RC oscillator by means of an external 8MHz reference clock applied to the XTALP pin with XTALN pin kept floating.
- The new functions **Sys_CalibrateUlpoByExtClk** and **Sys_CalibrateLfoByExtClk** allow calibrating 1 kHz and 32 kHz LFO RC oscillators by means of an external 8MHz reference clock applied to the XTALP pin with XTALN pin kept floating.
- The new functions **Accel_RawToGX** and **Accel_RawToGZ** can be used to estimate an acceleration change that corresponds to the measured change of ADC units [LSB]. The estimation is based on the factory calibration of the acceleration gain.

2.6.3 Changes

- The input parameter with PGA gain is removed from the functions **low_power_adc_read_pressure**. Now the function implements a raw low-power pressure measurements with calibrated PGA gain stored in the nvRAM at Melexis factory.
- The functions **Accel_ToAduX** and **Accel_ToAduZ** were renamed to the **Accel_GToRawX** and **Accel_GToRawZ** correspondingly. The former names are marked as deprecated.
- **LF_GAIN_SEL** supports 2 gain levels with the maximal **LF_AGC_TRIM** value configured in the premain code similarly for all profiles. The details are available in the description of the function **Lf_Init**.
- The profiles configurations in the *.mk files are cleaned from the unused parameters (e.g. the unsupported options for MCU clock are removed).

2.7 Version 1.10.0

The library supports chip variants with different **pressure ranges** (100-500 kPa, 100-900 kPa and 100-1400 kPa) and different **acceleration sensing** (X, Z, XZ-axes or without acceleration sensor). Three variants of input ranges are available for the options with a single sensitive acceleration axis. The required chip variant has to be selected in the application's Makefile by assigning the **PROFILE** variable by one of the profiles names listed below:

PROFILE	Description
DEFAULT	Alias for 91804CA_9BAR_XZ; should not be used for application
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91803BC_9BAR	BC: 100-900 kPa without acceleration sensor
91803BC_14BAR	BC: 100-1400 kPa without acceleration sensor
91803CA_5BAR	CA: 100-500 kPa without acceleration sensor
91803CA_9BAR	CA: 100-900 kPa without acceleration sensor
91803CA_14BAR	CA: 100-1400 kPa without acceleration sensor
91804BC_5BAR_X	BC: 100-500 kPa, X-acceleration sensor with symmetrical input range
91804BC_5BAR_X_NEG	BC: 100-500 kPa, X-acceleration sensor with asymm negative input range
91804BC_5BAR_X_POS	BC: 100-500 kPa, X-acceleration sensor with asymm positive input range
91804BC_5BAR_Z	BC: 100-500 kPa, Z-acceleration sensor with symmetrical input range
91804BC_5BAR_Z_NEG	BC: 100-500 kPa, Z-acceleration sensor with asymm negative input range
91804BC_5BAR_Z_POS	BC: 100-500 kPa, Z-acceleration sensor with asymm positive input range
91804BC_9BAR_X	BC: 100-900 kPa, X-acceleration sensor with symmetrical input range
91804BC_9BAR_X_NEG	BC: 100-900 kPa, X-acceleration sensor with asymm negative input range
91804BC_9BAR_X_POS	BC: 100-900 kPa, X-acceleration sensor with asymm positive input range
91804BC_9BAR_Z	BC: 100-900 kPa, Z-acceleration sensor with symmetrical input range
91804BC_9BAR_Z_NEG	BC: 100-900 kPa, Z-acceleration sensor with asymm negative input range
91804BC_9BAR_Z_POS	BC: 100-900 kPa, Z-acceleration sensor with asymm positive input range
91804BC_14BAR_Z	BC: 100-1400 kPa, Z-acceleration sensor with symmetrical input range
91804BC_14BAR_Z_NEG	BC: 100-1400 kPa, Z-acceleration sensor with asymm negative input range
91804BC_14BAR_Z_POS	BC: 100-1400 kPa, Z-acceleration sensor with asymm positive input range
91804CA_5BAR_X	CA: 100-500 kPa, X-acceleration sensor with symmetrical input range
91804CA_5BAR_X_NEG	CA: 100-500 kPa, X-acceleration sensor with asymm negative input range
91804CA_5BAR_X_POS	CA: 100-500 kPa, X-acceleration sensor with asymm positive input range
91804CA_5BAR_Z	CA: 100-500 kPa, Z-acceleration sensor with symmetrical input range
91804CA_5BAR_Z_NEG	CA: 100-500 kPa, Z-acceleration sensor with asymm negative input range
91804CA_5BAR_Z_POS	CA: 100-500 kPa, Z-acceleration sensor with asymm positive input range
91804CA_5BAR_XZ	CA: 100-500 kPa, XZ-acceleration sensor with symmetrical input range
91804CA_9BAR_X	CA: 100-900 kPa, X-acceleration sensor with symmetrical input range
91804CA_9BAR_X_NEG	CA: 100-900 kPa, X-acceleration sensor with asymm negative input range
91804CA_9BAR_X_POS	CA: 100-900 kPa, X-acceleration sensor with asymm positive input range
91804CA_9BAR_Z	CA: 100-900 kPa, Z-acceleration sensor with symmetrical input range
91804CA_9BAR_Z_NEG	CA: 100-900 kPa, Z-acceleration sensor with asymm negative input range
91804CA_9BAR_Z_POS	CA: 100-900 kPa, Z-acceleration sensor with asymm positive input range
91804CA_9BAR_XZ	CA: 100-900 kPa, XZ-acceleration sensor with symmetrical input range
91804CA_14BAR_X	CA: 100-1400 kPa, X-acceleration sensor with symmetrical input range
91804CA_14BAR_X_NEG	CA: 100-1400 kPa, X-acceleration sensor with asymm negative input range
91804CA_14BAR_X_POS	CA: 100-1400 kPa, X-acceleration sensor with asymm positive input range
91804CA_14BAR_Z	CA: 100-1400 kPa, Z-acceleration sensor with symmetrical input range
91804CA_14BAR_Z_NEG	CA: 100-1400 kPa, Z-acceleration sensor with asymm negative input range
91804CA_14BAR_Z_POS	CA: 100-1400 kPa, Z-acceleration sensor with asymm positive input range
91804CA_14BAR_XZ	CA: 100-1400 kPa, XZ-acceleration sensor with symmetrical input range

For example, to build the application with the 91804CA_9BAR_X profile, the Makefile should include the line:

```
PROFILE = 91804CA_9BAR_X
```


The DEFAULT profile is to simplify switching all library examples to any required profile, because all examples' makefiles include the line `PROFILE = DEFAULT`. By default the DEFAULT profile is an alias for the 91804CA_9BAR_XZ profile. It is not recommended to build an application with the DEFAULT profile.

2.7.1 Fixed issues

The functions **RfTx_Run**, **Sys_SleepWith** and **Sys_DeepSleepWith** were adjusted to run with active interrupts.

2.7.2 New features

- Added new profiles supporting asymmetrical acceleration input ranges (see above).
- Enabled gcc's section garbage collection (linker option **-gc-sections**) to remove unused library functions from the application binary file.
- Added new functions **Flash_GetSingleErrorCounter** and **Flash_ClearSingleErrorCounter** to get status and control single error counter (SEC) of the Flash memory.
- Added new functions **NvRam_IsDoubleErrorDetected** and **NvRam_ClearDoubleError** to get states and clear Double Error in the NVRAM memory.
- Added new function **LfRx_ReceivedHeader** to get 32-bits of received LF header.
- Added new function **Lf_GetState** to get the current state of LF RX module.
- Added new function **RfTx_RunFromBuf** that performs a completed RF transmission of the payload data stored in memory array. The function allows transmitting up to 255 bytes of payload (> RfTx FIFO size).
- Added new functions **Check_IsBadAccelZ** and **Check_IsBadAccelX** to diagnose Z and X axes of the acceleration sensor.
- Added new function **Check_IsBadPressure** to diagnose the pressure sensor.
- Added new functions **Sys_SetNormalLfo** and **Sys_SetUlpLfo** to switch between LFO modes.
- Added SW support of the I2C serial interface in master mode.
- Added new parameter **.extra** to **RfTxConfig_Type** (used by **RfTx_Init** function). The field **.extra** points to an additional RFTX configuration (**RfTxExtraConfig_Type**) that defines RFTX central frequency divider value (**RFTX_CENTER_FREQ_L**, **RFTX_CENTER_FREQ_H** and **RFTX_CENTER_FREQ_U**), XTAL capacitor configuration (**PLL_EN_XTAL_INT_CAP**) and RFTX mode selection (**RfTxPresetId_Type**) to specify the RFTX power mode, VPA bypass mode and RFTX frequency range.

The library provides a macro **RFTX_EXTRA_CONFIG** macro function to simplify the additional RFTX configuration. The macro includes three parameters, where the first one (**RFTX_FREQ_IN_KHZ**) defines the RFTX central frequency expressed in kHz that replaces three RFTX divider's values mentioned above. Second parameter (**PRESET_ID**) selects pre-set that specifies the RFTX power mode, VPA bypass mode and RFTX frequency range. And third parameter (**XTAL_CAP_ID**) configures capacitor load on the crystal (external capacitors or internal ones, see **XtalCap_Type** for the details). Please, see **app_biphase_encoding** example for the reference.

If parameter **.extra** is set to 0 (or not mentioned), the default extra configuration is fixed by the library:

- RFTX central frequency is set to 433.92 MHz (**RFTX_CENTER_FREQ_IN_KHZ** = 433920);
- Internal capacitance of the nodes **XTALP** and **XTALN** is configured with 6pF (**PLL_EN_XTAL_INT_CAP_ID** = 1);
- RFTX output power is set to 5 dBm for all library profiles supporting 5 or 9 bar pressure ranges and set to 8 dBm for other library profiles supporting 14 bar pressure range;
- VPA regulated mode is applied (**RF_VPA_BYPASS** is disabled).

2.7.3 Changes

- Reworked the source code to further comply with MISRA-C 2012 requirements.
- Inverted the error logic returned by the functions **Flash_PageWrite** and **Flash_SectorErase**.
Now Flash_PageWrite and Flash_SectorErase functions return true if operation failed and false if it is succeeded!
 As a result, the implemented logic matches the return values of other library functions: Sys_CalibrateHfo, Sys_CalibrateLfo, etc.
- Renamed file `tpmslib\_version.h` to `lib\_version.h` and renamed version defines as below

Old API	New API
TPMSLIB_VERSION_MAJOR	LIB_VERSION_MAJOR
TPMSLIB_VERSION_MINOR	LIB_VERSION_MINOR
TPMSLIB_VERSION_PATCH	LIB_VERSION_PATCH
TPMSLIB_VERSION_STRING	LIB_VERSION_STRING
TPMSLIB_GIT_HASH	LIB_GIT_HASH

- Atomic blocks syntax has been changed

Old API	New API
<pre>ENTER_SECTION(ATOMIC_KEEP_MODE) { /* code */ }</pre>	<pre>ENTER_SECTION(ATOMIC_KEEP_MODE); /* code */ EXIT_SECTION();</pre>

- Renamed folder `products/91804/` to `custom/`
- Renamed linker script template from 91804.ldt to symb-addr.ld
- Function **Adc_ReadVoltage** now uses two ADC samples (with different polarity) to improve measurement accuracy.
- Functions **Wakeup_Init**, **TempSensor_Init**, **Gpio_WriteOpenDrainPin**, **Gpio_ToggleOpenDrainPin**, **Itc_SwIntRequest**, **NvRam_RecallNoWait**, **NvRam_StoreNoWait**, **NvRam_IsBusy**, **Adc_ClearErrors**, and **Sys_IsWarmBoot** are marked as deprecated. It is not recommended to use them in new designs, as they can be removed from the next library releases. Now the deprecated functions are declared in the header file `deprecated.h`.
- The next header files become deprecated:
 abs_watchdog.h, accel.h, accel_hw_filter.h, adc.h, calc.h, gpio.h, itc.h, lf.h, nvram.h, nvram_map.h, phys.h, radio.h, rf.h, rftx_private.h, simple_timer.h, syslib.h, syslib_ext.h, temp_sensor.h, wakeup.h, wakeup_timer.h.
 Please use `sensorlib.h` instead.

2.7.4 Removed features

- Removed profiles: 91804BC_14BAR_XZ, 91804BC_5BAR_XZ and 91804BC_9BAR_XZ
- Removed `as\_field` wrapper in `NvRamCfg\_Type` for nvram_mlx object (file nvram_map.h). Now NVRAM fields can be accessed as below

Old API	New API
nvram_mlx.as_field.PRESS_PGA_GAIN	nvram_mlx.PRESS_PGA_GAIN

- Removed **.CYCLE_SLIP** parameter of the RfTxConfig_Type (used by **RfTx_Init** function) and RfTxStatus_Type (used by **RfTx_GetStatus** function). Removed corresponding **RfTx_PllClearCycleSlip** function.
- Removed wakeups of the Diag Interface from Wakeup_Enable, Wakeup_Disable and Wakeup_Init functions

- Makefile's variable **PRODUCT** is not needed anymore
- Simplified *.mk profile configuration:

1) Removed parameters:

ADC_AVERAGING_NUMBER_FOR_PRESSURE
 ADC_AVERAGING_NUMBER_FOR_VOLTAGE
 ADC_AVERAGING_NUMBER_FOR_TEMP
 ADC_AVERAGING_NUMBER_FOR_ACCEL

To implement an efficient averaging inside of any function from the left column of the table below, **it is a responsibility of application** to override the corresponding low level function at the middle column by executing a single measurement desired times with the next averaging of the resultant data. The single measurement is implemented by the functions at the right column of the table below:

Adc_ReadPressure	adc_ll_read_pressure	adc_ll_read_pressure_once
Adc_ReadVoltage	adc_ll_read_vbat	adc_ll_read_vbat_once
Adc_ReadTemperature	adc_ll_read_temperature	adc_ll_read_voltage_once
Phys_AccelerationX Phys_AccelerationZ	adc_ll_read_accel	adc_ll_read_accel_once
Adc_ReadGpio	adc_ll_read_gpio	adc_ll_read_voltage_once

See **adc_sens** and **adc_gpio** examples for the reference.

If the application doesn't override the low level function at the middle column above, the library assigns it to be equivalent to the function from the right column, and hence the corresponding library function at the left column will use **a single measurement by default. Melexis is responsible for this default option only!**

- 2) Removed **HAS_ULP_OSCILLATOR** parameter. The ULP mode is always enabled by default.
- 3) Removed **ACC_FILTER_BUFFER_DEPTH** parameter. Now 16 samples are used by default there.
- Removed the deprecated functions below:

Removed deprecated function	Suggested replacement
mlx16_disable_interrupts	builtin_mlx16_disable_interrupts
mlx16_set_priority	builtin_mlx16_set_priority
mlx16_get_status	builtin_mlx16_get_status
mlx16_set_status	builtin_mlx16_set_status
mlx16_enter_user_mode	builtin_mlx16_enter_user_mode
_fsb	builtin_mlx16_fsb
_sfb	builtin_mlx16_sfb
_fsb32	builtin_mlx16_fsb32
_sfb32	builtin_mlx16_sfb32
RfTx_Deinit	Radio_Disable
Sys_Halt	Sys_HaltWith(LF_DISABLED)
Sys_Stop	Sys_StopWith(LF_DISABLED)
Sys_Sleep	Sys_SleepWith(LF_DISABLED)
Sys_DeepSleep	Sys_DeepSleepWith(LF_DISABLED)
Sys_StopWithLfRx	Sys_StopWith(LF_RECEIVE)
Sys_SleepWithLfRx	Sys_SleepWith(LF_RECEIVE)
Sys_DeepSleepWithLfRx	Sys_DeepSleepWith(LF_RECEIVE)
Sys_SleepAfterRfTx	use RfTx_Run and Sys_SleepWith
Sys_DeepSleepAfterRfTx	use RfTx_Run and Sys_DeepSleepWith
Accel_DeinitHWFilter	Accel_DeinitFilter
Adc_ReadPVTWithGain	NA
Adc_ReadPressureWithGain	NA

2.7.5 Migration guide

In order to port an application firmware from the SW library versions 1.9.0 to the version 1.10.0 the firmware has to be adjusted according to all items listed in two sections above.

2.8 Version 1.9.0

This release improves library safety and stability, increases performance and simplifies user experience:

- Unified and simplified API for RFTX and Sleep functions
- Code change to increase compliance with MISRA-C 2012 requirements
- Reduced application building time:
 - Initial build time is reduces **by 50%**
 - Rebuild time of the application with LOADER is reduces **6 times**

Library is configurable to support chip variants with different **pressure sensors** (5, 9 and 14 bar) and **acceleration sensors** (X, Z or XZ-axes). Required variant is selected by assigning one of the profiles below to the **PROFILE** variable to in the application's Makefile.

PROFILE	Description
91803BC_5BAR	100-500 kPa without acceleration sensor
91803BC_9BAR	100-900 kPa without acceleration sensor
91803BC_14BAR	100-1400 kPa without acceleration sensor
91803CA_5BAR	CA: 100-500 kPa without acceleration sensor
91803CA_9BAR	CA: 100-900 kPa without acceleration sensor
91803CA_14BAR	CA: 100-1400 kPa without acceleration sensor
DEFAULT	alias for 91804CA_9BAR_XZ; should not be used for application
91804BC_5BAR_X	100-500 kPa, X-acceleration sensor
91804BC_5BAR_Z	100-500 kPa, Z-acceleration sensor
91804BC_5BAR_XZ	100-500 kPa, XZ-acceleration sensor
91804BC_9BAR_X	100-900 kPa, X-acceleration sensor
91804BC_9BAR_Z	100-900 kPa, Z- acceleration sensor
91804BC_9BAR_XZ	100-900 kPa, XZ-acceleration sensor
91804BC_14BAR_Z	100-1400 kPa, Z-acceleration sensor
91804BC_14BAR_XZ	100-1400 kPa, XZ-acceleration sensor
91804CA_5BAR_X	CA: 100-500 kPa, X-acceleration sensor
91804CA_5BAR_Z	CA: 100-500 kPa, Z-acceleration sensor
91804CA_5BAR_XZ	CA: 100-500 kPa, XZ-acceleration sensor
91804CA_9BAR_X	CA: 100-900 kPa, X-acceleration sensor
91804CA_9BAR_Z	CA: 100-900 kPa, Z- acceleration sensor
91804CA_9BAR_XZ	CA: 100-900 kPa, XZ-acceleration sensor
91804CA_14BAR_X	CA: 100-1400 kPa, X-acceleration sensor
91804CA_14BAR_Z	CA: 100-1400 kPa, Z-acceleration sensor
91804CA_14BAR_XZ	CA: 100-1400 kPa, XZ-acceleration sensor

Default configuration profile `PROFILE=DEFAULT` is an alias for the 91804CA_9BAR_XZ profile. To build library with different profile, change PROFILE variable in the Makefile's of the example. E.g. to build for the 91804CA_9BAR_X profile, use:

```
``` Makefile
PRODUCT=91804
PROFILE=91804CA_9BAR_X
#...
```

## 2.8.1 Fixed issues

- Fixed loader to avoid forcing the `.had_por` flag to 1 on exit. This caused incorrect status returned by the `Sys_GetResetReason` function.

## 2.8.2 New features

1) Added new examples:

### **diag\_accel**

- Demonstrates Acceleration diagnostic
- Located at code/examples/diag\_accel

### **diag\_pressure**

- Demonstrates pressure sensor diagnostic
- Located at code/examples/diag\_pressure

### **reference\_example**

- Provides a template that can be used as an example to start own TPMS application firmware design
- Located at code/examples/reference\_example

2) New functions in the library:

### **Lf\_Disable**

- stops and disables LF block

### **NvRam\_RestoreHfoTrimming**

### **NvRam\_RestoreLfoTrimming**

### **NvRam\_RestoreUlpoTrimming**

- restores factory calibration for HFO, LFO and ULPO oscillators

### **Radio\_Disable**

- stops and disables RF and LF radio interfaces
- this function should be used instead of deprecated `RfTx_DeInit`

### **RfTx\_Run**

- Runs to completion RF transmission in STOP mode (blocking call)
  - executes proper pre-charge procedure and then transmits using STOP mode
  - waits until transmission is fully finished and then exits
  - requires `ISR_RFTX_DONE` interrupt, therefore application SHALL NOT use `ISR_RFTX_DONE` for own purposes
  - **warning: LF RX is disabled inside this function and should be explicitly re-enabled on exit, if needed**
  - **function returns true if any error is detected during the execution**

```
...
 if (RfTx_Run()) {
 /* error detected */
 }
 /* else: TX succeeded */
...
```

### **Sys\_HaltWith**

### **Sys\_StopWith**

### **Sys\_SleepWith**

### **Sys\_DeepSleepWith**

- Enters the power down mode (halt, stop, sleep or deep sleep) with specified LF mode: `LF_DISABLED`, `LF_RECEIVE`, `LF_RSSI` or `LF_TRANSMIT`
- RFTX is explicitly disabled for `Sys_SleepWith` and `Sys_DeepSleepWith` functions
- RFTX is kept unchanged for `Sys_HaltWith` and `Sys_StopWith` functions

DEBUG\_PUT\_BYTE  
 DEBUG\_PUT\_WORD  
 DEBUG\_PUT\_WORD\_HEX  
 DEBUG\_PUT\_STRING  
 DEBUG\_PIN\_INIT  
 DEBUG\_PIN\_TOGGLE  
 DEBUG\_PIN\_PULSE  
 DEBUG\_ACK\_AWD

- The functions can be used to output debug information via serial interface (SW UART with a bit rate 115200 bps) or as a logic level pulses on debug pin that is assigned to GPIO0.
- By default, the debug functions are disabled and all their invocations are ignored and not included into the final application.
- To enable the debug functions, add `DDEBUG\_ENABLED` define to `CPPLAGS` variable in the Makefile.

### 2.8.3 Removed features

- 1) ITC\_RFTX\_DONE interrupt and WAKEUP\_RFTX\_DONE wakeup event **SHALL NOT be used by application**, as those will be handled by the library in RfTx\_Run function.
- 2) Function Sys\_EnterPowerDown has been removed.

One of the following functions can be used instead:

- Sys\_HaltWith
- Sys\_StopWith
- Sys\_SleepWith
- Sys\_DeepSleepWith

- 3) The following functions had been deprecated in previous library release and remove in this release

Removed deprecated functions / features	Suggested replacement in 1.9.0
Accel_InitHWFilterForXAxis	Accel_InitFilterForX
Accel_InitHWFilterForZAxis	Accel_InitFilterForZ
Accel_DeinitHWFilter	Accel_DeinitFilter
Accel_PickNext	Accel_PeekNext
Adc_ReadAccelerationX Adc_ReadAccelerationXRef Adc_ReadAccelerationXWithGain Adc_ReadAccelerationXRefWithGain Calc_AccelerationX	Use composite function instead: Phys_AccelerationX
Adc_ReadAccelerationZ Adc_ReadAccelerationZRef Adc_ReadAccelerationZWithGain Adc_ReadAccelerationZRefWithGain Calc_AccelerationZ	Use composite function instead: Phys_AccelerationZ
removed direct access to NVRAM parameters DIE_POS and DIE_LOT_WFR	Sys_GetChipId

### 2.8.4 Changes

- 1) Updated Makefiles in the examples to ease PROFILE, PROTECTIONKEY64bit and LOADER options selection.
- 2) New version of IO port HAL is included, which enables more optimization for the IO\_SET and IO\_GET functions.



3) Functions below are marked as deprecated and will be removed in the next release

Deprecated functions	Suggested replacement in 1.9.0
RfTx_DeInit	Radio_Disable  - note that it also disables LF interface
Sys_Halt Sys_Stop	Sys_HaltWith(LF_DISABLED) Sys_StopWith(LF_DISABLED)  - note that RFTX was explicitly disabled by old deprecated functions, while new functions keep the configured RFTX_MODE <b>unchanged</b>
Sys_Sleep Sys_DeepSleep	Sys_SleepWith(LF_DISABLED) Sys_DeepSleepWith(LF_DISABLED)
Sys_StopWithLFRx	Sys_StopWith(LF_RECEIVE)  - note that in old deprecated function the RFTX was disabled; new function keeps the configured RFTX_MODE <b>unchanged</b>
Sys_SleepWithLFRx Sys_DeepSleepWithLFRx	Sys_SleepWith(LF_RECEIVE) Sys_DeepSleepWith(LF_RECEIVE)
Sys_SleepAfterRfTx(lf_mode)	Sequence: RfTx_Run(); Sys_SleepWith(lf_mode);
Sys_DeepSleepAfterRfTx(lf_mode)	Sequence: RfTx_Run(); Sys_DeepSleepWith(lf_mode);

4) Optimized **loader** to reduce its size for the future features/improvements

5) Enhanced error reporting for the following **calibration** functions:

- RfTx\_Calibrate
- Sys\_CalibrateHfo
- Sys\_CalibrateLfo
- Sys\_CalibrateUlp0

From now on, an overflow in frequency counter is monitored and reported. Also status of XTAL start and PLL lock are monitored and reported.

As before, functions return `true` if calibration failed for some reason, including frequency function overflow:

```

...
 if (RfTx_Calibrate()) {
 /* error detected */
 }
 /* else: calibration succeeded */
...

```

## 2.9 Version 1.8.2

This is a patch release to fix a bug in the LF bootloader with a wrong stack allocation.

## 2.10 Version 1.8.1

This library release supports new chip revisions **91804CA**, **91803BC** and backward compatible with previous revision **91804BC**.

Library is configurable to support chip variants with different **pressure sensors** (5, 9 and 14 bar) and **acceleration sensors** (X, Z or XZ-axes). Required variant is selected by assigning one of the profiles below to the **PROFILE** variable to in the application's Makefile.

PROFILE	Description
<b>91803BC_5BAR</b>	<b>[new]</b> 100-500 kPa without acceleration sensor
<b>91803BC_9BAR</b>	<b>[new]</b> 100-900 kPa without acceleration sensor
<b>91803BC_14BAR</b>	<b>[new]</b> 100-1400 kPa without acceleration sensor
<b>91803CA_5BAR</b>	<b>[new]</b> CA: 100-500 kPa without acceleration sensor
<b>91803CA_9BAR</b>	<b>[new]</b> CA: 100-900 kPa without acceleration sensor
<b>91803CA_14BAR</b>	<b>[new]</b> CA: 100-1400 kPa without acceleration sensor
DEFAULT	alias for 91804CA_9BAR_XZ; <b>should not be used for application</b>
91804BC_5BAR_X	100-500 kPa, X-acceleration sensor
<b>91804BC_5BAR_Z</b>	<b>[new]</b> 100-500 kPa, Z-acceleration sensor
<b>91804BC_5BAR_XZ</b>	<b>[new]</b> 100-500 kPa, XZ-acceleration sensor
91804BC_9BAR_X	100-900 kPa, X-acceleration sensor
91804BC_9BAR_Z	100-900 kPa, Z- acceleration sensor
91804BC_9BAR_XZ	100-900 kPa, XZ-acceleration sensor
91804BC_14BAR_Z	100-1400 kPa, Z-acceleration sensor
91804BC_14BAR_XZ	100-1400 kPa, XZ-acceleration sensor
<b>91804CA_5BAR_X</b>	<b>[new]</b> CA: 100-500 kPa, X-acceleration sensor
<b>91804CA_5BAR_Z</b>	<b>[new]</b> CA: 100-500 kPa, Z-acceleration sensor
<b>91804CA_5BAR_XZ</b>	<b>[new]</b> CA: 100-500 kPa, XZ-acceleration sensor
<b>91804CA_9BAR_X</b>	<b>[new]</b> CA: 100-900 kPa, X-acceleration sensor
<b>91804CA_9BAR_Z</b>	<b>[new]</b> CA: 100-900 kPa, Z- acceleration sensor
<b>91804CA_9BAR_XZ</b>	<b>[new]</b> CA: 100-900 kPa, XZ-acceleration sensor
<b>91804CA_14BAR_X</b>	<b>[new]</b> CA: 100-1400 kPa, X-acceleration sensor
<b>91804CA_14BAR_Z</b>	<b>[new]</b> CA: 100-1400 kPa, Z-acceleration sensor
<b>91804CA_14BAR_XZ</b>	<b>[new]</b> CA: 100-1400 kPa, XZ-acceleration sensor

Default configuration profile `PROFILE=DEFAULT` is an alias for the 91804CA\_9BAR\_XZ profile. To build library with different profile, change PROFILE variable in the Makefile's of the example. E.g. to build for the 91804CA\_9BAR\_X profile, use:

```

``` Makefile
PRODUCT=91804
PROFILE=91804CA_9BAR_X
#...
```

```

### 2.10.1 Fixed issues

Not applicable

### 2.10.2 New features

- New devices: Added configuration profiles for **91803BC** devices (profiles 91803BC\_5BAR, 91803BC\_9BAR, 91803BC\_14BAR), **91803CA** devices (profiles 91803CA\_5BAR, 91803CA\_9BAR, 91803CA\_14BAR), **91804BC** devices (profiles 91804CA\_5BAR\_Z and 91804CA\_5BAR\_XZ) and **91804CA** devices (profiles 91804CA\_5BAR\_X, 91804CA\_5BAR\_Z, 91804CA\_5BAR\_XZ, 91804CA\_9BAR\_X, 91804CA\_9BAR\_Z, 91804CA\_9BAR\_XZ, 91804CA\_14BAR\_X, 91804CA\_14BAR\_Z and 91804CA\_14BAR\_XZ)
- Profile: Added possibility to set **RFTX\_CENTER\_FREQ\_IN\_KHZ** in two ranges: **[300000..330000]**kHz or **[420000..450000]**kHz
- The LF boot loader: Communication format was optimized to support up to **64-byte LF data telegrams** that eventually reduces programming time
- Acceleration: Added new functions to convert from the acceleration physical units to ADC units. There are two function for X and Z axis respectively: **Accel\_ToAduX** and **Accel\_ToAduZ**. These functions can be used to convert physical value of acceleration threshold (expressed in [g] multiplied by 10) into ADC units to be further used to compare with the raw acceleration values obtained by the uncompensated acceleration measurements (e.g. with configurable Low Pass Hardware Filter).
- Acceleration: Added functions **Phys\_AccelerationX** and **Phys\_AccelerationZ** which return acceleration along X and Z axis measured in physical units: [g] multiplied by 10. These functions are going to replace the sequence of the following functions which are now deprecated:

```
/* old API [deprecated] */
int16_t raw = Adc_ReadAccelerationX();
int16_t ref = Adc_ReadAccelerationXRef();
int16_t phys_accel = Calc_AccelerationX(raw, ref);

/* new API */
int16_t phys_accel = Phys_AccelerationX();
```

### 2.10.3 API changes

- 1) For 91804CA devices, the following functions used for 91804BC are deprecated: **Accel\_InitHWFilterForXAxis**, **Accel\_InitHWFilterForZAxis** and **Accel\_DeinitHWFilter**. Use corresponding functions from the table below.

**[!]** Please note, that time base for the timing parameters in these new CA functions is changed to **0.1 milliseconds** (was 0.125 milliseconds for deprecated BC functions)

| 91804BC functions<br>(deprecated for 91804CA)  | 91804CA functions                                |
|------------------------------------------------|--------------------------------------------------|
| Accel_InitHWFilterForXAxis [time base 0.125ms] | Accel_InitFilterForX [time base <b>0.100ms</b> ] |
| Accel_InitHWFilterForZAxis [time base 0.125ms] | Accel_InitFilterForZ [time base <b>0.100ms</b> ] |
| Accel_DeinitHWFilter                           | Accel_DeinitFilter                               |

- 2) Direct access to chip ID stored in NVRAM is **deprecated!** Use new function **Sys\_GetChipId** from syslib.h instead.  
E.g. replace deprecated code below by the new one, using Sys\_GetChipId() call.

```
/* deprecated code */
#define Chip_ID_0 ((uint8_t)nvram_mlx.as_port.DIE_POS)
...
RfTx_FifoWriteByte(Chip_ID_0);

/* recommended code */
RfTx_FifoWriteByte(Sys_GetChipId().as_byte[0]);
```

3) The following acceleration measurement function are deprecated and shall be replaced by the new ones

| Old API (deprecated)                                                    | New API            |
|-------------------------------------------------------------------------|--------------------|
| Accel_PickNext                                                          | Accel_PeekNext     |
| Adc_ReadAccelerationX<br>Adc_ReadAccelerationXRef<br>Calc_AccelerationX | Phys_AccelerationX |
| Adc_ReadAccelerationZ<br>Adc_ReadAccelerationZRef<br>Calc_AccelerationZ | Phys_AccelerationZ |

```

/* old API [deprecated] */
int16_t raw = Adc_ReadAccelerationX();
int16_t ref = Adc_ReadAccelerationXRef();
int16_t phys_accel = Calc_AccelerationX(raw, ref);

/* new API */
int16_t phys_accel = Phys_AccelerationX();

```

#### 2.10.4 Library Changes / Optimization

- Profiles: changed the name of default profile from 91804.mk to DEFAULT.mk
- Acceleration:
  - Optimized HW filter functions for 91804CA devices
  - Optimized acceleration fine offset compensation for 91804CA devices
  - Extended calibration (10-bits) for offset compensation for 91804CA devices
- Acceleration calculation functions:
  - For 91804BC devices, calculation functions **Calc\_AccelerationX**, **Calc\_AccelerationZ** etc. try to use **calibrated gain**, if such calibration is available in the NVRAM (systematically implemented in production since 12th of July 2017). Otherwise, the fixed constant gain is used for the backward compatibility. The 91804CA devices always use calibrated gain.
- RFTX: Optimized RFTX pre-charge time
- LF: Auto-gain trimming (AGC\_TRIM) is set to 2 for 91804CA devices and to 3 for 91804BC
- Library version is stored at the address 0x6028 in the following format:
  - TPMSLIB\_GIT\_HASH ; git hash of lib snapshot; 4-bytes, LSByte first
  - TPMSLIB\_VERSION\_MAJOR ; 1-byte
  - TPMSLIB\_VERSION\_MINOR ; 1-byte
  - TPMSLIB\_VERSION\_PATCH ; 1-byte
  - record version ; 1-byte

#### 2.10.5 Removed features

Not applicable